

Andres Felipe Pineda

Senior Backend Engineer · Payments & Fintech Platforms

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Backend engineer with 13+ years building and shipping production software, with experience across payments, e-commerce, and data-heavy platforms. I like refactoring codebases toward clean domain models and care about tests, tooling, and leaving systems easier to reason about than I found them. PhD in Systems & Informatics.

EXPERIENCE

Senior Backend Engineer — TipHaus

Apr 2024 – Present · Remote

SaaS platform for tip management & distribution for U.S. restaurants and hospitality — Laravel 12, PHP 8.4, MySQL, Redis, GraphQL, Livewire, Docker.

- Designed and built **NextGen ETA**, an Event-Sourcing payouts engine (aggregates, projectors, reactors) processing instant tip payments to employees, and migrated every legacy payout to event-sourced aggregates with zero service downtime.
- Integrated and operated **6+ banking & payment providers** — Astra Finance, Checkbook, Plaid, Upwardli, RapidPay, Synctera, ZumRails — including webhooks, transfer reconciliation, KYB/KYC, and business onboarding.
- Led **provider migrations moving real user funds**: transferred user balances Synctera→Upwardli and mass-migrated organizations Checkbook→Astra, with readiness-audit commands and batch validation.
- Fixed **critical security vulnerabilities**: authentication bypass via unsigned JWT, account takeover during registration, and QR-payment amount tampering that allowed draining funds.
- Led the monolith's refactor to **Domain-Driven Design** (Financial, Distributions, Reports, Integrations) and modernized the stack: Laravel 10→12, PHP 8.2→8.4, Passport v13, PHPStan 2.
- Built the QR-tipping registration flow and wallet pre-funding reconciliation, plus enterprise features — linking multiple users to one Astra account across organizations and cross-org configuration copy.
- Developed internal **Livewire admin tooling**: event-sourcing dashboard, webhook simulator, migration boards, and operational "playgrounds" for support.
- Raised financial-domain test coverage to **90%**, fixing bugs surfaced along the way, and migrated POS integrations (Clover OAuth v1→v2).
- Integrated AI-assisted development (Claude Code, Cursor) — building a custom agent harness with skills, MCP servers, rules, specialized agents (security auditor, financial), Laravel Boost, and spec-driven development for complex money-math work.

Lead Developer — autoFlow

Dec 2021 – Mar 2024

Multi-tenant SaaS for print-production workflow automation — Laravel 11, Livewire 3, PostgreSQL, AngularJS.

- Designed and built from scratch a **multi-tenant SaaS platform** automating print-production workflows (Laravel 11, Livewire 3, PostgreSQL, AngularJS) as lead developer.

- Architected the system with **Domain-Driven Design** (~15 bounded contexts — Jobs, Quotes, Proofs, Forms, Transfers, Enrollments), an Actions pattern, ViewModels, and per-module role / license access control.
- Built the core business modules: job management with configurable checkpoints, quote sign-off with multi-tenant invitations, versioned print-proof review with PDF annotations and a mobile-optimized viewer, a secure file-transfer system (uTransfer), and a dynamic form builder with custom field types and dropdowns populated from REST APIs via JSONPath.
- Integrated the **Enfocus ecosystem** — bidirectional communication with Switch (Send, Receive, Process, Review, ServerAgent) and PDF processing via PitStop Server CLI (preflight, AI-assisted bleed correction, cross-platform thumbnail generation).
- Implemented integrations with **5+ industry MIS/ERP systems** (EPMS via SOAP; PACE, Nubium, PSV via REST; Printavo→PSV automated billing) plus QuickBooks, behind a configurable, admin-managed "data sources" layer with export, import, and conflict resolution.
- Created self-contained **native installers for macOS and Windows** (nginx + PHP + PostgreSQL 17, NSSM services, Laravel scheduler and queues) with in-place updates, automatic DB backup/restore, health checks and diagnostic bundles — removing the need for on-site technical intervention.
- Built real-time infrastructure with Server-Sent Events; diagnosed complex cross-platform production issues (PHP-FPM/libpq GSSAPI segfaults, PostgreSQL version mismatches, Windows path escaping, opcache on deploys); and kept quality high with Pest, PHPStan/Larastan, Pint pre-commit hooks, and Laravel Pennant feature flags.
- Integrated AI-assisted development (Claude Code, Cursor) — adding custom skills, MCP servers, rules, Laravel Boost, and spec-driven development.

Backend Engineer — L3Funding

Jan 2019 – Nov 2021

Fintech CRM for business funding (merchant cash advance)

- Core full-stack engineer on a fintech CRM for business funding, working across deal management, lead routing, underwriting, finances, and reporting.
- Built a reusable multi-step wizard engine powering a multi-brand online funding application with **Plaid** bank linking, **DocuSign** embedded signing, S3 document uploads, round-robin rep assignment, and concurrency-safe submission that auto-creates deals in the CRM.
- Built a **funder matchmaking engine** evaluating deals against configurable funder criteria (rules, operators, live preview) to automate deal-to-funder routing, plus an AI-assisted risk-profiling feature integrating DataMerch.
- Designed a communications data lake ingesting email, calls, and SMS (Gmail, RingCentral, Instantly) with ClickHouse indexing, Redis/Horizon queueing, cross-vendor identity resolution, and a real-time operator dashboard.
- Built **Cortex**, an AI sales-rep agent — LLM orchestrator with guard pipeline, provider-agnostic models, tool drivers, contextual memory, dry-run mode, and an inspector UI.
- Led major upgrades — Laravel 10→11→12, Livewire 2→3→4 — and migrated the suite to parallel **Pest** (~47% faster); established the DDD architecture (domains, internal packages, actions pattern) and authored the team's AI-agent engineering guidelines.

Technical Interviewer — Toptal

Jul 2018 – Dec 2018

- Conducted technical interviews for full-stack developer candidates.

Full-Stack Engineer — Blazetrak

Jan 2016 – Jun 2018

- Refactored the legacy platform from `mysql_connect` to PDO and Doctrine 2, and moved to an MVC pattern.
- Built a mini-framework (routing, helpers, security, mail, repositories, console) via Composer to standardize platform functionality.
- Migrated asset management to Bower / Gulp / Elixir; moved uploads from DropZone to fineUploader & CameraTag with Amazon S3 storage.
- Improved performance by refactoring SQL queries, adding indexes and APC; automated platform activities and email campaigns with cron jobs.
- Refactored payments to DirectPayment and ExpressCheckout (PayPal), adding Subuno for fraud screening.

Software Engineer — Infinity Health Tech

Jul 2015 – Dec 2015

- Refactored legacy code from unstructured to an OOP approach; set up workflow with NPM, Gulp and Composer.
- Moved assets to AWS S3 and managed deployment across dev, staging, demo and production.
- Implemented modules for user, patient, doctor and test management.

Lead Developer — TICademia

Jan 2015 – Aug 2021

Production edtech platform for learning math & programming with gamification.

- Led the migration of TICademia to **Laravel 8 / PHP 8** as lead author of the repository, coordinating with two other developers via GitHub.
- Redesigned and modularized the backend: refactored authentication, registration, purchases/payments, certificates, courses, and routing across ~60 Eloquent models and a service layer (gamification, competitive programming, analytics, real-time).
- Built product features including an achievements system, notifications, a student friendship/social network, and certificate generation.
- Hardened the authentication flow: fixed password hashing, enforced email confirmation before login, and added password-reset notifications.
- Modernized the frontend build pipeline, migrating from Gulp to Webpack (Laravel Mix 6), and developed reusable JavaScript components.
- Worked with a production stack including MySQL, Redis, Pusher (real-time), Mailgun, social OAuth (Laravel Socialite), and DigitalOcean.

Backend Engineer — Q Customer Intelligence (Lumiary)

Jul 2014 – Dec 2014

- Integrated the platform with Zendesk, MailChimp, Stripe, Bronto, Lob and BigCommerce via API, storing data in MongoDB.
- Built Node.js APIs and modules for a new schema (invitations, segmenting, accounts, authorization, dashboards, settings, orders).
- Added Elasticsearch API support; managed application logs with AWS CloudWatch and cached queries with AWS ElastiCache.
- Created API-request test cases and fixed application bugs.

Front-End Developer — Globant

Jan 2014 – Jun 2014

- Built plugins with infuse.js and delivered pixel-perfect proofs of concept for several American newspapers.
- Consumed a JSP API to add front-end functionality; maintained legacy JSP and resolved cross-browser / mobile issues.

Backend Engineer — ColumbusSoft

Jul 2013 – Dec 2013

- Built a Laravel module integrating the Amazon API to extract and publish product data; implemented the checkout process via Amazon API in PHP.
- Designed and built interfaces for an online vitamin shop with Foundation; maintained internal projects (scheduling, billing, tasks) and migrated projects Laravel 3→4.

Lead Developer — MySmarHealth

Jul 2012 – Jun 2013

- Defined the architecture of an ophthalmology clinic-management platform; built the core with Laravel 4 and jQuery.
- Migrated Laravel 4→5 and jQuery→AngularJS; oversaw code review, QA and deployment with Git and Bitbucket.
- Managed hosting and domains, and set up the full developer workflow with Composer and Gulp.

Backend Engineer — Optilodging

Jan 2012 – Jun 2012

- Built a booking calendar with jQuery promises and a hotel-billing module in CodeIgniter.
- Programmed a generic guest-data export for national-police validation; designed the MySQL database and built responsive Bootstrap interfaces.

TECHNOLOGIES & SKILLS

Backend	PHP · Laravel · Livewire · Inertia · Composer · Node.js · Python · Java	Frontend	JavaScript (ES6) · Vue.js · React · Alpine.js · TailwindCSS · Bootstrap · SASS
Databases	MySQL · PostgreSQL · Redis · MongoDB · Oracle / PL-SQL	Search	MeiliSearch · Elasticsearch · Algolia · Sphinx
Infra / DevOps	Docker · Nginx · Laravel Forge · AWS · GCP · DigitalOcean · GitHub · CI/CD	Testing	Pest · PHPUnit · Postman · k6 · Larastan (PHPStan) · Xdebug · Blackfire
Monitoring	Sentry · New Relic · Datadog	Payments	Stripe · PayPal · Plaid · Braintree · Avalara · Astra · Checkbook · Synctera · ZumRails
AI & Tools	Claude Code · Cursor · Gemini · PHPStorm · VS Code	Practices	DDD · TDD · Event Sourcing · Vector embeddings · Composer · Webpack

EDUCATION

Ph.D. in Systems & Informatics	National University of Colombia
M.Sc. in Systems & Informatics	National University of Colombia
B.Eng. in Systems & Informatics	National University of Colombia

SELECTED PUBLICATIONS

- A Framework for Automated Formative Assessment in Mathematics Courses — *IEEE Access*, vol. 8, pp. 30152–30159.
- A hybrid approach for composing groups in collaborative learning contexts — *Heliyon*, vol. 7, issue 6, e07249.
- SMoT+: Extending the SMoT Algorithm for Discovering Stops in Nested Sites — *Computing and Informatics*, vol. 33, no. 2.

LANGUAGES

English	Advanced (C1)
Spanish	Native
Portuguese	Intermediate

CONFERENCES

- Laracon US — Nashville, 2023
- Laracon US — Denver, 2025